

One size-two atom suffices for the quartic tracial moment problem on $Y^2 = 1$

Candidate full solution packet for arXiv:2001.11614
Agent `agent_lane.03` Model GPT5.6

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Status. `full_solution_likely_valid`; subject to expert review and a broader novelty search.

1 Source problem

Bhardwaj and Zalar study normalized bivariate quartic tracial moment sequences and representing measures by pairs of real symmetric matrices, using normalized matrix trace [1]. Their Conjecture 1, on PDF page 12, asks whether, under the column relation $Y^2 = 1$, the following are equivalent: existence of an arbitrary representing measure; existence of a measure using only scalar and size-two atoms; and existence of a measure of type $(m, 1)$, i.e. with exactly one size-two atom.

ously feasible. Hence $\mathcal{M}_2$ does not admit a representing measure.

4. $\mathcal{M}_2$ OF RANK 6 WITH RELATION $Y^2 = 1$

The main result of this section, see Theorem 4.1 below, is that moment matrices $\mathcal{M}_2$ generated by the atoms (X, Y) of size 3 satisfying $Y^2 = I_3$ can always be represented with atoms of size at most 2. Moreover, if we consider a single atom of size 3, then a single atom of size 2 suffices.

Theorem 4.1. *Let β be a moment sequence with a nc moment matrix $\mathcal{M}_2$ satisfying the column relation $Y^2 = 1$. Then the following are equivalent:*

- (1) $\mathcal{M}_2$ admits a measure of type (m_1, m_2, m_3) , $m_1, m_2, m_3 \in \mathbb{N} \cup \{0\}$.
- (2) $\mathcal{M}_2$ admits a measure of type (m_1, m_2) , $m_1, m_2 \in \mathbb{N} \cup \{0\}$.

Moreover, if $m_3 = 1$ in (1) the $m_2 = 1$ in (2).

The proof is constructive and can be seen as the first step toward proving the following conjecture:

Conjecture 1. *Let β be a moment sequence with a moment matrix $\mathcal{M}_2$ satisfying the column relation $Y^2 = 1$. Then the following are equivalent:*

- (1) $\mathcal{M}_2$ admits a measure.
- (2) $\mathcal{M}_2$ admits a measure of type (m_1, m_2) , $m_1, m_2 \in \mathbb{N}$.
- (3) $\mathcal{M}_2$ admits a measure of type $(m, 1)$, $m \in \mathbb{N}$.

Let $\beta^{(4)}$ be a truncated moment sequence and $\mathcal{M}_2$ its moment matrix. The notations $\Delta(\beta^{(4)})$ and $\Delta(\mathcal{M}_2)$ will both denote the difference

The preceding Theorem 4.1 in [1] proves this when the original measure has no atoms larger than size three. The theorem below removes that restriction and proves the full conjecture.

2 Definitions and a scalar moment fact

A finite representing measure can be encoded as a finite-dimensional real $*$ -algebra

$$\mathcal{A} = \bigoplus_{j=1}^N M_{n_j}(\mathbb{R})$$

with the tracial state

$$\tau(T) = \sum_{j=1}^N \rho_j \frac{1}{n_j} \operatorname{tr}(T_j), \quad \rho_j > 0, \quad \sum_j \rho_j = 1.$$

The given moments are $\beta_w = \tau(w(X, Y))$. Conversely, such a direct sum is precisely a finite tracial representing measure.

We use the following standard degree-four Hamburger fact (equivalently, the one-variable flat moment theorem; see, for example, [2]).

Lemma 1 (Scalar flat completion). *Let $s_0, s_1, s_2, s_3 \in \mathbb{R}$ and put*

$$H = \begin{pmatrix} s_0 & s_1 \\ s_1 & s_2 \end{pmatrix}, \quad g = \begin{pmatrix} s_2 \\ s_3 \end{pmatrix}.$$

If $H \succ 0$, then setting $s_4 = g^T H^{-1} g$ produces a positive semidefinite flat Hankel matrix

$$\begin{pmatrix} H & g \\ g^T & s_4 \end{pmatrix},$$

and $(s_0, \dots, s_4)$ has a representing measure on $\mathbb{R}$ with at most two atoms. The same conclusion holds with $H^\dagger$ when $H \succeq 0$ and $g \in \operatorname{ran} H$; in rank one the representing measure is a point mass.

3 Proof intuition

Diagonalizing the involution Y splits X into two diagonal corners and one off-diagonal block:

$$X = \begin{pmatrix} A & B \\ B^* & C \end{pmatrix}, \quad Y = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

All failure of commutativity is measured, through degree four, by B . We choose a single size-two atom whose off-diagonal entry absorbs the total energy $\tau(BB^*)$, while its diagonal entries are the BB^* - and B^*B -weighted means of A and C . This choice cancels every mixed term through degree three. What remains on the two signs of Y is ordinary one-variable moment data.

The apparent obstruction is the fourth moment: making the size-two atom too light forces its off-diagonal entry to be large. The exact minimum fourth moments of the two scalar residuals are Schur complements. Orthogonal projection in the two corner L^2 spaces turns their sum into a one-variable function G . Two applications of Cauchy–Schwarz and one elementary two-variable inequality show $G \geq 0$ at an explicit parameter. Since $G \rightarrow -\infty$ at the other end, continuity gives a parameter with $G = 0$; both scalar completions are then flat. This simultaneously handles every atom in the original measure and leaves exactly one size-two atom.

4 The theorem

Theorem 1. *Let $\beta = (\beta_w)_{|w| \leq 4}$ be a normalized bivariate real tracial moment sequence whose moment matrix satisfies the column relation $Y^2 = 1$. If β has a representing measure, then it has a representing measure of type $(m, 1)$. If β is noncommutative, one may take $m \leq 4$. Consequently all three conditions in Conjecture 1 of [1] are equivalent.*

Proof. Encode an arbitrary finite representing measure by $(\mathcal{A}, \tau, X, Y)$ as above. The column relation is inherited atomwise: indeed,

$$0 = \tau((Y^2 - 1)^2) = \sum_j \rho_j \frac{1}{n_j} \text{tr}((Y_j^2 - I_{n_j})^2),$$

and every summand is nonnegative because Y_j is symmetric. Thus $Y_j^2 = I_{n_j}$ for every atom, and $Y^2 = 1$ in $\mathcal{A}$. Put

$$e_+ = \frac{1+Y}{2}, \quad e_- = \frac{1-Y}{2},$$

and, relative to these two corners, write

$$X = \begin{pmatrix} A & B \\ B^* & C \end{pmatrix}.$$

All traces below are taken with τ ; block formulas are interpreted componentwise in the direct sum.

Step 1: the moment identities

Set

$$D = BB^*, \quad D' = B^*B, \quad w = \tau(D) = \tau(D'), \quad S = \tau(D^2) = \tau((D')^2).$$

For $0 \leq k \leq 3$ let

$$p_k = \frac{\beta_{X^k} + \beta_{X^k Y}}{2} = \tau(e_+ X^k), \quad q_k = \frac{\beta_{X^k} - \beta_{X^k Y}}{2} = \tau(e_- X^k).$$

Direct block multiplication gives

$$\begin{aligned} p_0 &= \tau(e_+), & q_0 &= \tau(e_-), \\ p_1 &= \tau(A), & q_1 &= \tau(C), \\ p_2 &= \tau(A^2) + w, & q_2 &= \tau(C^2) + w, \\ p_3 &= \tau(A^3) + 2\tau(AD) + \tau(CD'), & q_3 &= \tau(C^3) + \tau(AD) + 2\tau(CD'). \end{aligned}$$

Moreover,

$$\beta_{X^4} = \tau(A^4) + \tau(C^4) + 4\{\tau(A^2 D) + \tau(C^2 D') + \tau(ABCB^*)\} + 2S, \quad (1)$$

$$\Delta := \beta_{X^2} - \beta_{XYXY} = 4w. \quad (2)$$

If $w = 0$, then $B = 0$ in the tracial L^2 norm and X commutes with Y . Spectral measures of A and C give a scalar representing measure. Replacing part of any scalar atom by the identical scalar pair on a two-dimensional space gives a representation of type $(m, 1)$ if that exact format is desired. Hence assume $w > 0$, so also $S > 0$.

Step 2: the unique natural size-two atom

Define the weighted means

$$a = \frac{\tau(AD)}{w}, \quad c = \frac{\tau(CD')}{w}.$$

For a parameter $\lambda > 0$, let

$$\tilde{X} = \begin{pmatrix} a & b \\ b & c \end{pmatrix}, \quad \tilde{Y} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad b^2 = \frac{w}{\lambda},$$

and give this size-two atom density 2λ . Since normalized trace on M_2 is one half of ordinary trace, the atom contributes λ times the ordinary two-dimensional trace. In particular its contribution to Δ is $4\lambda b^2 = 4w$, exactly (2).

The plus-corner contributions through degree three are

$$\lambda, \quad \lambda a, \quad \lambda(a^2 + b^2), \quad \lambda\{a^3 + b^2(2a + c)\},$$

and the minus-corner formulas interchange a and c . By the definitions of a, c and $\lambda b^2 = w$, subtraction therefore leaves

$$r_k^+ = \tau(A^k) - \lambda a^k, \quad qquadr_k^- = \tau(C^k) - \lambda c^k, \quad qquad 0 \leq k \leq 3. \quad (3)$$

Thus the remaining problem is scalar on each sign of Y .

Step 3: positivity and exact scalar fourth moments

Work first in the plus corner, with inner product $\langle R, T \rangle = \tau(RT)$. Let $L_A = \text{span}\{e_+, A\}$, let Q_A be the orthogonal projection of A^2 onto L_A , and put

$$R_A = A^2 - Q_A, \quad E_A = \|R_A\|_2^2.$$

Evaluation of an affine-linear polynomial at a is represented on L_A by the projection of D/w . Write

$$\frac{D}{w} = P_{L_A}\left(\frac{D}{w}\right) + z_A, \quad qquad \kappa_A = \left\|P_{L_A}\left(\frac{D}{w}\right)\right\|_2^2, \quad qquad s_A^2 = \|z_A\|_2^2 = \frac{S}{w^2} - \kappa_A.$$

If $Q_A = \alpha e_+ + \gamma A$, define

$$h_A = \alpha + \gamma a, \quad qquad \delta_A = h_A - a^2, \quad qquad V_A = \tau((A - ae_+)^2 D).$$

The same definitions, with C, D', e_- , give $R_C, E_C, z_C, \kappa_C, s_C, h_C, \delta_C, V_C$.

Cauchy-Schwarz shows that for every affine-linear ℓ ,

$$|\ell(a)|^2 = \left| \tau\left(\ell(A)\frac{D}{w}\right) \right|^2 \leq \tau(\ell(A)^2) \frac{S}{w^2}.$$

Consequently, whenever

$$\frac{1}{\lambda} \geq \frac{S}{w^2}, \quad (4)$$

both order-one Hankel matrices associated to (3) are positive semidefinite. Also $w^2 \leq S\tau(e_+)$ and $w^2 \leq S\tau(e_-)$, so (4) implies $\lambda \leq \min\{\tau(e_+), \tau(e_-)\}$ and $2\lambda \leq 1$.

Assume first that the base plus Hankel matrix is nonsingular. A Sherman–Morrison calculation says that the least fourth moment completing $(r_0^+, \dots, r_3^+)$ is

$$q_A(\lambda) = \tau(Q_A^2) - \lambda a^4 + \frac{\lambda \delta_A^2}{1 - \lambda \kappa_A}. \quad (5)$$

Indeed, the base order-one Hankel matrix is the Gram matrix of e_+, A ; its cubic cross-vector represents Q_A , and subtracting $\lambda(1, a)^T(1, a)$ gives (5). The identical formula holds for $q_C(\lambda)$. If a base Hankel matrix has rank one, the corresponding corner operator is scalar in $L^2(\tau)$ and the same formula holds by continuity, with the zero quotient interpreted as zero. At a singular endpoint the necessary range condition is likewise the limiting one.

Two identities will control the apparent poles. Orthogonality of R_A to L_A gives

$$\delta_A = \frac{V_A}{w} - \langle R_A, z_A \rangle, \quad \delta_C = \frac{V_C}{w} - \langle R_C, z_C \rangle. \quad (6)$$

For example, $\tau(R_A D/w) = \tau(A^2 D/w) - h_A = V_A/w - \delta_A$, and the left side is $\langle R_A, z_A \rangle$.

Step 4: the fourth-moment surplus

Put

$$T = V_A + V_C + \tau((A - ae_+)B(C - ce_-)B^*).$$

The ordinary trace of $\tilde{X}^4$ is

$$a^4 + c^4 + 4b^2(a^2 + ac + c^2) + 2b^4.$$

Using (1), (5), and $\lambda b^2 = w$, the amount of fourth moment left after paying for the size-two atom and the two minimal scalar completions is exactly

$$\begin{aligned} G(\lambda) &= E_A + E_C + 4T + 2S - \frac{2w^2}{\lambda} \\ &\quad - \frac{\lambda \delta_A^2}{1 - \lambda \kappa_A} - \frac{\lambda \delta_C^2}{1 - \lambda \kappa_C}. \end{aligned} \quad (7)$$

Write

$$\frac{1}{\lambda} = \frac{S}{w^2} + v, \quad v \geq 0.$$

Then (7) becomes

$$G(v) = E_A + E_C + 4T - 2w^2v - \frac{\delta_A^2}{s_A^2 + v} - \frac{\delta_C^2}{s_C^2 + v}. \quad (8)$$

By (6) and Cauchy–Schwarz in the direct sums $L^2 \oplus \mathbb{R}$,

$$\frac{\delta_A^2}{s_A^2 + v} \leq E_A + \frac{V_A^2}{w^2v}, \quad \frac{\delta_C^2}{s_C^2 + v} \leq E_C + \frac{V_C^2}{w^2v}. \quad (9)$$

Here $v > 0$; the zero case is read as a limit.

Let $U = (A - ae_+)B$ and $V = B(C - ce_-)$. Then $V_A = \|U\|_2^2$, $V_C = \|V\|_2^2$, and the last term in T is $\langle U, V \rangle$. Hence

$$T \geq V_A + V_C - \sqrt{V_A V_C}.$$

For all $x, y \geq 0$,

$$4(x + y - \sqrt{xy}) \geq 2\sqrt{2}\sqrt{x^2 + y^2}; \quad (10)$$

after squaring, the difference is $2(\sqrt{x} - \sqrt{y})^4$.

If $V_A^2 + V_C^2 > 0$, choose

$$v_0 = \frac{\sqrt{V_A^2 + V_C^2}}{\sqrt{2}, w^2}.$$

Equations (8)–(10) give

$$G(v_0) \geq 4T - \left(2w^2v_0 + \frac{V_A^2 + V_C^2}{w^2v_0}\right) = 4T - 2\sqrt{2}\sqrt{V_A^2 + V_C^2} \geq 0.$$

If $V_A = V_C = 0$, use $v_0 = 0$. Then (6) gives $\delta_A^2/s_A^2 \leq E_A$ and $\delta_C^2/s_C^2 \leq E_C$ (with the usual zero convention), so again $G(v_0) \geq 0$.

On the other hand $G(v) \rightarrow -\infty$ as $v \rightarrow \infty$. It follows by continuity that there is $\hat{v} \geq v_0$ such that $G(\hat{v}) = 0$. Choose

$$\lambda = \left(\frac{S}{w^2} + \hat{v}\right)^{-1}.$$

For this choice, the remaining fourth moment is exactly $q_A(\lambda) + q_C(\lambda)$.

Step 5: construct the scalar atoms

Give the plus residual in (3) fourth moment $q_A(\lambda)$ and the minus residual fourth moment $q_C(\lambda)$. Lemma 1 produces a scalar measure on the line $y = 1$ and one on the line $y = -1$, each with at most two atoms. Together with the single size-two atom $(\tilde{X}, \tilde{Y})$ of density 2λ , these measures match p_k, q_k for $0 \leq k \leq 3$, match β_{X^4} , and match Δ . Cyclicity and $Y^2 = 1$ reduce every word of degree at most four to these data, so the resulting measure represents all of β .

Thus a noncommutative β has a measure of type $(m, 1)$ with $m \leq 4$. The implications from type $(m, 1)$ to a measure using atoms of sizes at most two and then to an arbitrary representing measure are immediate. This proves all equivalences in the source conjecture. $\square$

5 Verification and limitations

The accompanying script `code/verify_random_reduction.py` checks the block identities, residual cubic matching, exact surplus formula, and its proved lower bound on 5,000 seeded random finite tracial direct sums. This is a regression test only; the proof above is analytic.

The delicate review points are: the coefficient 4 in (1); the Sherman–Morrison term in (5); and the limiting convention when a corner Hankel matrix has rank one. In the latter case the corner operator is scalar in $L^2(\tau)$, so its residual is a point mass and the formulas are literal limits of the nonsingular case.

A bounded novelty check was performed on 9 August 2026. Exact local-corpus and arXiv/web searches used arXiv:2001.11614, the paper title and authors, and the phrases “ $Y^2 = 1$ ”, “type $(m, 1)$ ”, and “atoms of size at most 2”. They found the source and earlier adjacent work, but no later resolution. This does not establish novelty; a broader expert literature search remains necessary.

6 Human-review recommendation

High priority. The argument is short once expressed in a finite tracial direct sum, it answers the exact conjecture, and it strengthens the conclusion to at most four scalar atoms in the noncommutative case. A reviewer should recompute (1) and (7), then check the two singular endpoint cases against Lemma 1.

References

- [1] A. Bhardwaj and A. Zalar, *The tracial moment problem on quadratic varieties*, J. Math. Anal. Appl. 498 (2021), 124936; arXiv:2001.11614.
- [2] N. I. Akhiezer, *The Classical Moment Problem and Some Related Questions in Analysis*, Oliver & Boyd, 1965.